EXPERIMENTAL EXAMINATION

December 10, 2005

Maximum points for problem is 20

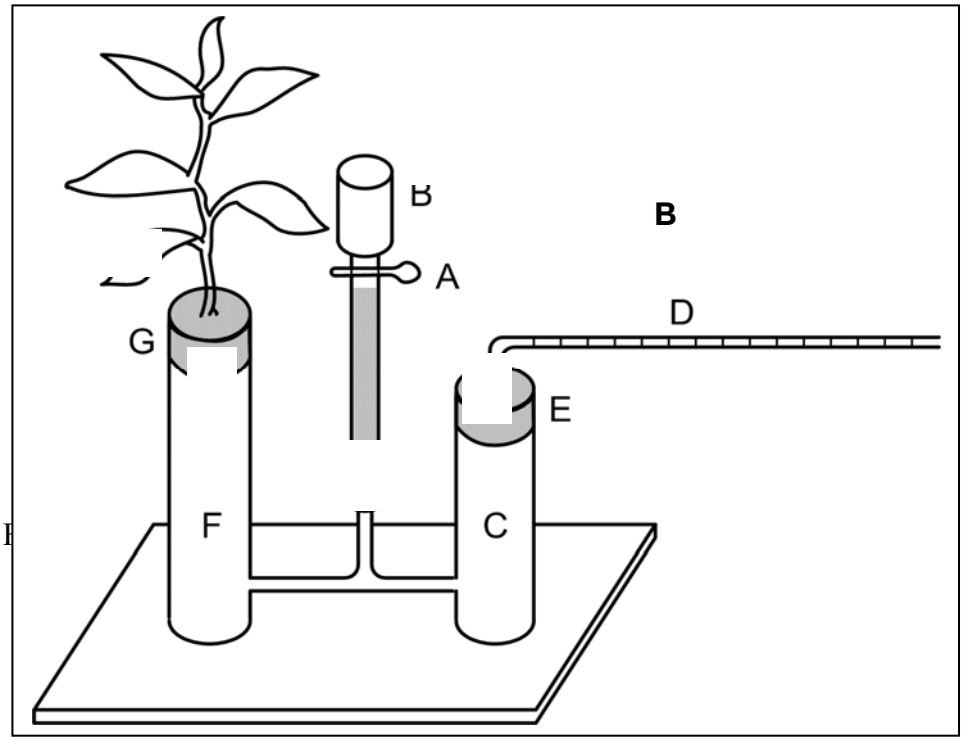
A. Introduction

The following experiment deals with transpiration phenomena which usually occurs in plants, i.e. water loss as water vapor from living tissues of the aerial parts of the plants. A laboratory apparatus known as "potometer" is usually used to quantitatively measure the transpiration rate of such plants, employing the amount of water consumed by the plant for a given period of time.

Due to the differences in the structure of plant tissues and in the its related physiological processes, the transpiration rate of one plant may differ considerably from the other. Moreover, several factors influencing transpiration rates, such as the atmospheric temperature, pressure, and humidity, as well as light intensity and wind speed have been well documented.

In this experiment, a simple laboratory made potometer, as schematically shown in Figure 1, will be used to quantify the influence of those factors on the transpiration rate of soybean (*Glycine max*) as a test plant.

The potometer consists of "water reservoir A", made of a capillary connected H-shaped glass tube and a valved funnel, and a capillary arm gauge - B is connected to the water reservoir with a rubber stopper. The test plant should be inserted into the water reservoir using a rubber support.



B. Objectives

B.1 General Objective

To demonstrate experimentally the transpiration process in a soybean test plant.

B.2 Specific Objectives

a. To measure quantitatively factors influencing transpiration rate of the test plant, such as light intensity and wind speed.
b. To recognize the influence of chemicals added to the media on the transpiration rate of the test plant.
c. To check participant knowledge on the physiological process involved in plant transpiration.

C. Apparatus and Materials

C.1 *Apparatus :*

1.	Potometer	1 set
2.	Stop watch	1 unit
3.	Circulating fan	1 set
4.	Tray	1 unit

C.2 *Materials :*

1.	Soybean plant	2 pots (1 plant per pot)
2.	Water	1 bottle (500 mL)
3.	Sugar solution of 2.5 % (w/w)	1 bottle (500 mL)
4.	Vaseline grease	1 tube
5.	Tissue paper	1 box
6.	Cutter	1 unit

D. Experiment

D.1 General Instruction

In the following experiments, you have to measure transpiration rates under different conditions e.g. light intensity, wind speed, and media. Experimentally, data are collected by measuring the water uptake due to the pulling force developed by transpiration. Read instructions carefully before doing the experiments. In each experiment, there will be questions related to the particular experiment. Answer all questions on the answer sheets provided.

D.2 Preparation steps

The following preparation steps should be carried out first, before doing any experiment.

1. Fix the capillary arm gauge B into the “water reservoir A” as shown in Figure 1.

2. Through the valved funnel, carefully fill up the "water reservoir A" of the potometer with water.
3. Prepare the test plant by cutting the soybean plant in the pot about 1 cm above the soil level, then fix the cut plant into the water reservoir of the potometer through the hole of rubber support provided. Be sure that all the cutting part of the plant has been immersed in the water of the reservoir!
4. You may use vaseline grease especially on the capillary arm gauge and the cut plant to avoid any leakage.

Experiment-I : Effects of Light on Transpiration Rate (5.5 points)

I-1 Set the water in the capillary arm gauge in full position (at the end of the arm). Be sure that no air bubble is present in the whole system of the potometer, otherwise serious difficulties will arise in subsequent observations. In the case air bubbles show up in your system, slightly open the rubber support and put in additional water to release the air.

I-2 Allow the system (for 15 minutes) for acclimatization. Under room light condition, record the length (in millimeters) of water column shifted (L) in the capillary arm gauge after 2, 5, and 10 minutes of transpiration process (1 point). Plot L (in millimeters) versus t (in minutes) from your data obtained in this particular experiment (1.5 points).

I-3 Similarly, after setting the water in the capillary arm gauge in full position, under "light off" condition (the test plant is shielded with a black cover available), record the length of water column shifted in the capillary arm gauge after 2, 5, and 10 minutes of transpiration process (1 points). Plot L (in millimeters) versus t (in minutes) from your data obtained in this particular experiment (1.5 points).

I-4 Based on the plots obtained in step I-2 and step I-3, choose one correct answer of the conclusion concerning the effects of light on the transpiration rates by crossing the letter of the correct answer on the answer sheet provided (0.5 point).

Experiment-II : Effects of Wind on Transpiration Rate (3 points)

II-1 This experiment is the continuation of experiment-I, finished previously. First of all, uncover the test plant. Then place and switch on circulating fan provided at a distance about 50 cm from the test plant in the potometer. Set the water in the capillary arm gauge at full position. Be sure that no air bubble is present in the whole system of the potometer.

II-2 Under the affect of "wind blowing" and room lighting, record the length (in millimeters) of water column shifted (L) in the capillary arm gauge after 2, 5, and 10 minutes of transpiration process (1 point). Plot L (in millimeters) versus t (in minutes) from your data obtained in this particular experiment (1.5 points).

II-3 Based on the plots obtained in step I-2 and those obtained in this experiment, choose one correct answer of the conclusion concerning the effects of wind on the transpiration rates by crossing the letter of the correct answer on the answer provided (0.5 point).

Experiment-III : Effects of Medium on Transpiration Rate (3 points)

III-1 Before conducting this particular experiment, you have to prepare an experimental set up using a different medium of transpiration, as follows: Pour out completely the water content in the potometer used previously. Through the valved funnel, carefully fill up the "water reservoir A" of the potometer with a 2.5 % (w/w) aqueous sugar solution available. Be sure that all the cut part of the plant has been immersed in the aqueous sugar solution of the reservoir!

III-2 Set the aqueous sugar solution in the capillary arm gauge at full position. Be sure that no air bubble is present in the whole system of the potometer, <u>otherwise</u> serious difficulties will arise in subsequent observations.

III-3 Under room light condition, record the length (in millimeters) of aqueous sugar solution column shifted (L) in the capillary arm gauge after 2, 5, and 10 minutes of transpiration process (1 point). Plot L (in millimeters) versus t (in minutes) from your data collected in this particular experiment (1.5 points).

III-4 Based on the plots obtained in step I-2 and those obtained in this experiment, choose one correct answer of the conclusion concerning the effects of medium on the transpiration rates by crossing the letter of the correct answer on the answer sheet provided (0.5 point).

E. *A. Questions*

B. You are provided with four possible answers. Read the question very carefully and then choose one correct answer by crossing one of the answers provided on the answer sheet. It should be noted that there is only one correct answer for each question.

1. The following data are observed in a transpirasion experiment.

Time (minute)	Water displacement (mm)
t_1	l_1
t_2	l_2
t_3	l_3

Assuming that the inner diameter of the capillary arm gauge is d mm, then the water volume taken up by the plant in a period between t_2 and t_3 would be (0.75 points):

A. $0.25\ \pi d^2(l_3 - l_2)$

B. $0.50\ \pi d^2(l_3 - l_2)$

C. $0.75\ \pi d^2(l_3 - l_2)$

D. $1.00\ \pi d^2(l_3 - l_2)$

2. Assuming that the water column displacement in the capillary arm gauge is linearly related to the transpiration time, then the calculated transpirasion rate of the plant, based on the data in question 1, would be (0.75 points):

A. $0.25\ \pi d^2(l_3 - l_2)/(t_3 - t_1)$

B. $0.25\ \pi d^2(l_3 - l_2)/(t_3 - t_2)$

C. $0.50\ \pi d^2(l_3 - l_1)/(t_3 - t_1)$

D. $0.50\ \pi d^2(l_3 - l_2)/(t_3 - t_2)$

3. When transpiration experiments of soybean were conducted in two different conditions, i.e. pure water and water containing 2.5 % (w/w) of sugar, the following observation could be expected (1 point).

A. No difference in the transpiration rate was observed with the two conditions applied.

B. The transpiration rates obtained in experiments using pure water medium were significantly lower than that using the water medium containing sugar.

C. The transpiration rates obtained in experiments using pure water medium were significantly higher than that using the water medium containing sugar.

D. No definite pattern in transpiration rate was observed with the two conditions applied.

4. Dealing with aqueous sugar solution, the following statement is *correct* (0.5 point).
   A. The density and viscosity of sugar solution are lower than those of water.
   B. The density and viscosity of sugar solution are higher than those of water.
   C. The density of sugar solution is higher than that of water, but the viscosity of an aqueous sugar solution is lower than that of water.
   D. The density of an aqueous sugar solution is lower than that of water, but the viscosity of sugar solution is higher than that of water.

5. In order to prepare a 2.5 % (w/w) aqueous sugar solution from 5.0 grams of sugar, the following amount of water should be added (0.5 point):
   A. 48.75 grams
   B. 97.50 grams
   C. 195.00 grams
   D. 390.00 grams

6. Among the formulas listed below, the chemical formula representing sugar obtained from sugar cane is (0.5 point):
   A. C_6H_5OH
   B. $C_6H_{12}O_6$
   C. $C_{12}H_{24}O_{12}$
   D. $C_{12}H_{22}O_{11}$

7. Both melting point and boiling point of substances may often be predicted based on their chemical formulas, molecular structures, and sometimes on their physical appearances. Concerning the melting point and the boiling point of sugar, the following statement is *correct* (0.5 point).

   A. Both the melting and boiling points of sugar are lower than those of water.
   B. Both the melting and boiling points of sugar are higher than those of water.
   C. Only the melting point of sugar is higher than that of water.
   D. Only the boiling point of sugar is higher than that of water.

8. Based on the involvement of heat observed experimentally, it could be concluded that the transpiration of soybean plant is (0.5 point):
   A. an endothermic process.
   B. an exothermic process.
   C. an adiabatic process.
   D. an isothermic process.

9. During your experiment of plant transpiration, water evaporation process occurring on the leaves of the soybean test plant, may be represented by the following chemical equation (0.5 point):
   A. $H_2O\ (s) \rightarrow H_2O\ (g)$
   B. $H_2O\ (g) \rightarrow H_2O\ (l)$
   C. $H_2O\ (l) \rightarrow H_2O\ (s)$
   D. $H_2O\ (l) \rightarrow H_2O\ (g)$

10. In the presence of light, both photosynthesis and transpiration are taking place in the chlorophilic parts of living plants, including in soybean leaves. One of the following chemical equations describes the photosynthesis (0.5 point):
   A. $2\ H_2\ (g) + O_2\ (g) \rightarrow 2\ H_2O\ (l)$
   B. $H_2O\ (l) + CO_2\ (g) \rightarrow$ carbohydrates $[C_n(H_2O)_m] + O_2\ (g)$
   C. $H_2O\ (l) + CO_2\ (g) \rightarrow H_2CO_3\ (l)$
   D. carbohydrates $[C_n(H_2O)_m] + O_2\ (g) \rightarrow CO_2\ (g) + H_2O\ (l)$

11. In principle, the transpiration investigated is due to (0.5 point):
   A. water pressure in the roots.
   B. water tension in the stem.
   C. absorption of water by the roots.
   D. evaporation of water from the leaves.

12. Transpiration mostly occurs from the leaves through their multitude of tiny holes called stomata. The stomata are responsible for (0.5 point):
   A. translocation of nutrients.
   B. releasing growth hormones.
   C. regulating water loss.
   D. transporting minerals.

13. Various environmental factors, directly influencing the opening and closing of the stomata, affect transpiration rates. The opening and closing of the stomata are controlled by (0.5 point):
    A. sieve tube member.
    B. guard cells.
    C. mesophyll.
    D. root hairs.

14. The presence of light is an important factor regulating plant transpiration in order to prevent plants from losing water. The involvement of photosynthesis in the transpiration process may cause stomata to (0.5 point):
    A. close, therefore the transpiration rate increases.
    B. close, therefore the transpiration rate decreases.
    C. open, therefore the transpiration rate increases.
    D. open, therefore the transpiration rate decreases.

15. Each of the following statements is related to factors affecting the rate of transpiration, *except* (0.5 point):
    A. Water evaporation is faster at higher temperatures.
    B. Transpiration increases as wind speed increases.
    C. Transpiration decreases when water supply is insufficient.
    D. Transpiration increases as humidity increases.